# HOLOTRADE

## An Exchange for Compute

Node-second settlement with explicit energy, workload-state, health, capability, locality, and basket adjustments

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*Many compute products still quote a rectangle of capacity for a span of time. Sub-second microVM startup makes finer metering useful, but not costless. Once cold-start cost is itemised, energy interval, workload state, health, capability, locality and basket topology can each be disclosed rather than flattened into one rate.*

*This paper is about what becomes priceable.*

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Built on the $W(3,3)$–$E_8$ substrate programme

`github.com/wilcompute/Holotrade`



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**How to read the evidence tags.** PROVED means an exact finite claim with a repository witness or exhaustive check. MEASURED means an observed property of the implemented prototype or a tool output such as synthesis; it does not mean production-market or physical-hardware evidence. MODELLED means a simulation output, policy coefficient or estimate. PUBLISHED means an external result and is accompanied by a citation. A passing regression test establishes that code follows its specification; it is not, by itself, evidence that an economic or physical model describes the world.

## Abstract

Holotrade is a research prototype for quoting and settling compute at node-second granularity while exposing energy, workload-state, utilisation, health, capability and locality adjustments separately. It combines three artifacts: a six-factor quote engine with an explicit cost floor, a seeded fleet simulation, and a table-free adjacency primitive based on the 40-point collinearity graph of $W(3,3)$. The finite graph facts and Boolean circuit equivalences are exact within their stated encodings. The fleet, prices, "genomes", wear, order flow and recursive fabric are models, not observations from a production venue. In a paired 64-seed experiment with 220 simulated nodes, the two-sided demand rule reduced terminal utilisation dispersion in all 64 runs; this establishes behaviour of the model, not longer hardware life or market efficacy. No physical photonic substrate or live exchange exists.

## Contributions and evidence taxonomy

| Tag | Supports | Does not support |
|---|---|---|
| PROVED | Exhaustive finite checks, explicit witnesses and SAT equivalence for the encoded circuits | Physical performance, economic adoption or an unstated generalisation beyond the checked object |
| MEASURED | Prototype behaviour and reproducible tool output | Production telemetry, customer outcomes or placed-and-routed hardware |
| MODELLED | Seeded simulations, policy rules, coefficients and conditional estimates | Empirical causal conclusions |
| PUBLISHED | A result in the cited primary source | Claims outside that source's assumptions or measurement environment |

The concrete contributions are: (i) an auditable quote decomposition and node-second settlement prototype; (ii) an exact implementation of the $W(3,3)$ adjacency graph and a separately scoped recursive address-distance model; and (iii) a software/RTL admission-policy artifact with explicit refusal reasons. The sections below keep those contributions separate from proposed market, security and hardware deployments.

# Contents

# 1 The thesis, in one page

Compute is increasingly traded as standardized capacity, while many products still display a fixed hourly-equivalent rate for a broadly specified machine. That convention suppresses information in four directions at once.

1. **It destroys the energy signal.** Wholesale electricity prices move every five minutes in ERCOT's real-time market and can be negative [8]. An hourly quote cannot expose that interval-level input. A node-second meter can associate each second with the grid-price interval that contains it; it does not create a new one-second wholesale price.
2. **It destroys the machine's history.** Two identical servers that have run different work may carry different caches, checkpoints, tuned kernels or scheduling state. Whether that state improves a future job is an empirical question. Holotrade represents the hypothesis explicitly so it can be tested rather than assuming that the physical chassis itself has learned.
3. **It destroys the wear signal.** A heavily used machine may be accumulating temperature excursions and usage-dependent wear, while a cold machine still draws standby power and ages on the calendar. A single rate does not expose either signal; the prototype models them separately, without claiming a validated lifetime effect.
4. **It destroys the shape.** Forty machines scattered across a datacentre and forty machines connected as one low-diameter block can have the same count but different usable connectivity. Holotrade therefore computes a separate basket-level topology score rather than treating it as a node property.

Holotrade's node quote exposes all four, together with capability and locality. The prototype displays an hourly-equivalent rate so buyers can compare familiar prices, then prorates that rate to the second at settlement:

$$P^{\mathrm{hr}} = P_0 \times E \times G \times D \times H \times Q \times L$$

hourly-equivalent quote; settlement rate is $P^{\mathrm{hr}}/3{,}600$ per node-second

The unit that settles is the **node-second**. The current model uses a 171.5 ms cold-start baseline, informed by the Firecracker microVM architecture [1]. MODELLED A fresh boot therefore consumes 17.15% of a one-second allocation and leaves 82.85% for useful work; boot overhead falls below 1% only once useful runtime exceeds roughly 17 seconds. Holotrade charges that cold start once and separately rather than concealing it in an hourly rounding rule.

**The paired simulation result motivating the balancer**

The demand term $D$ is *two-sided*: it charges a premium above a target utilisation band and offers a discount below it. Across 64 paired seeds, with 220 simulated nodes and 500 one-minute steps (8.333 simulated hours per arm), mean terminal utilisation Gini was **0.0645846** with the balancer on (95% CI [0.0604334, 0.0687358]) and **0.1610963** with it off (95% CI [0.1545541, 0.1676386]). Mean paired difference was **0.0965118** (95% CI [0.0930282,

> 0.0999953]); mean paired relative reduction was **0.6033545** (95% CI [0.5889816, 0.6177275]). All 64 paired runs improved. MODELLED
>
> This is a reproducible property of the stated demand-response model. It is not a measurement of a live market, temperature cycling, failure rate or hardware lifetime. Those consequences require telemetry and a separately validated wear model.

Everything else in this paper follows from taking those two moves seriously: a sub-second unit, and a price with observable parts.

# 2 What is actually being sold

## 2.1 The case against the hour

Many incumbent products quote node-hours. The hour is a commercial display and reservation convention, not a unit of computation. Faster provisioning makes finer settlement useful, but it does not eliminate metering, reconciliation, minimum-charge or cold-start costs.

**IN PLAIN ENGLISH: What a virtual machine is**

A physical server is one piece of hardware. A *virtual machine* is a software-created illusion of a whole separate computer running on top of it, so that one physical server can be rented to several customers at once without them being able to see or interfere with each other. The software that creates the illusion is called a *hypervisor*. Creating a traditional virtual machine involves booting a complete operating system inside the illusion, which is why it takes minutes.

**IN PLAIN ENGLISH: What a microVM is, and why it changes the economics**

A *microVM* is a stripped-down virtual machine: a real hypervisor boundary, a real separate operating-system kernel, but with everything unnecessary for one specific job removed — no virtual graphics card, no virtual sound card, no legacy device emulation. It gets the same security isolation as a full virtual machine with a reduced device model. Amazon's *Firecracker* is the best-known implementation and underlies AWS serverless products [1].

The prototype uses **171.5 ms** as its cold-start baseline. MODELLED Firecracker's published design and performance evaluation establish the relevance of sub-second microVM startup, but this paper does not present the 171.5 ms value as a new measurement or as portable across storage, kernel and host configurations [1]. Three design consequences follow.

1. **The hour has no physical justification.** A one-second billing slice carries 17.15% startup overhead in the worst case where every second is a fresh boot. The fraction is below 1% only beyond roughly 17 seconds of useful runtime. This supports itemised cold-start charging and second-level metering; it does not make startup free.

2. **Per-second energy pricing becomes possible rather than visible.** ERCOT publishes real-time prices on five-minute intervals [8]. Holotrade meters execution by the second and associates each metered second with its containing price interval; it does not interpolate a one-second wholesale price.

3. **The startup cost becomes visible instead of smeared.** Holotrade charges the boot *once, explicitly, as its own line*, rather than folding it into the rate. A customer running many short jobs can therefore see exactly what their fan-out pattern costs them — which is information the hourly model structurally cannot provide.

## 2.2 Three objects, not one

| Layer | Object | What it is |
|---|---|---|
| ASSET | the **node** | A durable machine with an identity, a modeled workload-state record, and a health record. This is what you own, lease, or lend into the pool. |
| CONTRACT | the **execution plan** | A document specifying intended software artefacts, permissions and a validity window. The prototype demonstrates an integrity seal and admission predicates; production digital signatures and attestation are future work. |
| UNIT | the **node-second** | What actually settles. Each second is associated with the wholesale-price interval in force when it was drawn. |

**IN PLAIN ENGLISH: What an execution plan is, and why it matters commercially**

Think of it as a specified work order. Before any machine starts, the plan states: here is the intended software digest, here are the network grants, and here is the validity window. A production version would sign that canonical content using a standard digital-signature scheme such as those specified by FIPS 186-5 [7], persist replay state, and bind admission to remote attestation. The present browser prototype does not implement those controls.

Commercially this separates a capacity reservation from a specification of intended execution. A signature authenticates that specification; it does not by itself prove what ran. Transferability would additionally require assignment rules, attestation, counterparty controls and jurisdiction-specific legal terms.

## 2.3 Why isolation is a precondition for the market existing at all

A two-sided marketplace requires that an owner of idle machines be willing to run a stranger's code on them. Without hardware-enforced isolation, "list your spare capacity" is a request to be compromised, and the supply side never materialises.

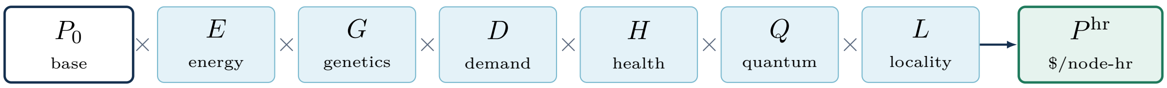
The deployment model Holotrade proposes gives each job its own kernel under a microVM hypervisor, deny-by-default network grants, verified filesystem images, remote attestation and an externally anchored audit log. Firecracker supports the reduced-device microVM boundary [1]; the remaining controls are requirements, not features established by this prototype.

**IN PLAIN ENGLISH: What a hash chain is**

Each entry in a correctly implemented hash chain commits to its content and the previous entry. Verification must recompute every digest, and the chain head must be retained or externally anchored; otherwise an attacker can rewrite the suffix or the whole chain. The current demo checks continuity and deletion of a middle entry only. It is an integrity-seal demonstration, not a production tamper-proof log.

# 3 The price, term by term

The formula is multiplicative rather than additive. That keeps each adjustment on its own displayed line and permits separate bounds. It does *not* make the inputs statistically or causally independent: utilisation, workload state, health and energy can be correlated. An explicit floor is applied after the multiplication, and a failed capability predicate makes the quote inadmissible rather than infinite.




each term shown separately; hourly equivalent settles pro rata by the second

## 3.1 $E$ — energy

**IN PLAIN ENGLISH**

Datacentres buy electricity on markets whose real-time settlement prices change by interval. ERCOT's real-time market uses five-minute settlement intervals and can publish negative or scarcity prices [8]. $E$ maps the current interval's price into a bounded quote adjustment.

$E$ is deliberately *not* linear in the grid price. A node should not become worthless when the grid goes negative, nor unsellable when it spikes. Holotrade currently applies a power law with exponent 0.55 to a positive-clamped ratio against the site's baseline, then clamps the multiplier to $[0.62, 2.40]$. MODELLED These are policy parameters, not estimated operator behaviour. Negative prices map to the lower clamp. PUE enters the separately displayed energy-cost floor, not this multiplier.

**IN PLAIN ENGLISH: What PUE is**

Power Usage Effectiveness. For every watt that reaches a computer chip, a datacentre spends additional watts on cooling, power conversion and lighting. PUE is the ratio of total facility power to computing power. A PUE of 1.0 would mean no facility overhead. In this prototype it multiplies the modeled energy cost used in the quote floor.

## 3.2 $G$ — genetics

This is the term that makes Holotrade a different product rather than a better price list.

**IN PLAIN ENGLISH**

When a machine runs the same class of work repeatedly, the software layer on it accumulates useful statistical structure — cached compilations, tuned kernel selections, warm model checkpoints, learned scheduling parameters. The industry already exploits this deliberately: "warm starts" and "checkpoint merging" are standard practice in machine-learning operations.

The testable hypothesis is that two physically identical servers carrying different reusable software state may deliver different time-to-result on a specified future workload. Holotrade calls that versioned, copyable software state the node's *genome*; the name does not imply that the chassis itself learns or that improvement is guaranteed.

$G$ is a model score built from three logged quantities. Their predictive value must be validated out of sample before $G$ can support a production premium:

- **Specialisation** on *this specific* workload class — not a general capability score. A node with excellent memory-bandwidth characteristics gets no credit for it on a workload that is compute-bound.
- **Realised fitness**: the completion record. Jobs finished versus jobs failed, weighted by experience depth. Never a manufacturer's specification.
- **Provenance depth**: lineage depth, treated here as a prior rather than proof of quality.

**The caveat that goes on every lease quote**

The simulation evolves a leased core toward the workload class it receives. MODELLED A production lease would need to define ownership, customer-data isolation, permitted updates, rollback and an out-of-sample benchmark. The quote therefore displays projected model drift as a scenario, not a measured 90-day forecast.

### 3.3 $D$ — demand and wear: the balancer

**IN PLAIN ENGLISH**

The obvious way to price demand is one-sided: a busy machine costs more. That is what surge pricing does, and it leaves the cold half of your fleet earning nothing while still aging, still drawing standby power, and still due for the same maintenance visit.

Holotrade charges a premium above a target band *and offers a discount below it.* In the prototype this is a demand-allocation policy. Whether its discount is cheaper than alternative operations is an empirical question.

**What the utilisation result does and does not say**

Temperature excursions can contribute to package fatigue, while sustained voltage, temperature and current contribute other wear mechanisms. The present simulation tracks utilisation and a modeled health trajectory; it does not measure junction temperature or count thermal cycles.

Lower cross-sectional utilisation Gini means work is distributed more evenly at the observation time. It does not imply lower temporal cycling, lower energy, longer life or de-clustered service events. Those are hypotheses for a telemetry study using temperature traces, failures and maintenance records.

**IN PLAIN ENGLISH: What a Gini coefficient is**

A single number between 0 and 1 measuring how unevenly something is distributed. Economists use it for income inequality. Here it measures how unevenly work is distributed across the fleet: 0 would mean every machine carries exactly the same load, 1 would mean one machine does everything while the rest sit idle. Lower is better.

**Paired 64-seed simulation**

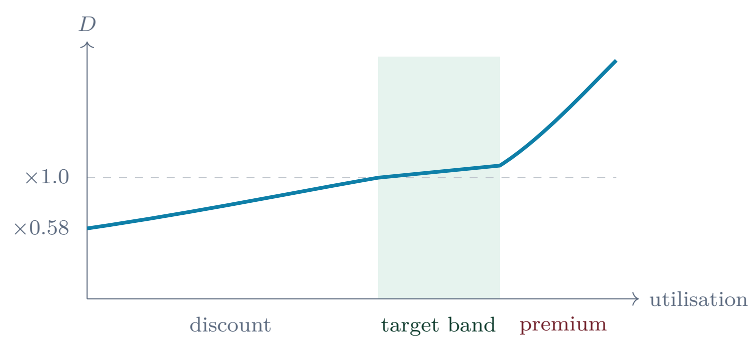
| | Mean | 95% CI | Paired outcome |
|---|---|---|---|
| Balancer off | 0.1610963 | [0.1545541, 0.1676386] | |
| Balancer on | **0.0645846** | [0.0604334, 0.0687358] | 64/64 lower |
| Paired difference | 0.0965118 | [0.0930282, 0.0999953] | off minus on |
| Relative reduction | 0.6033545 | [0.5889816, 0.6177275] | paired by seed |

Each pair uses the same seed, 220 nodes and 500 one-minute steps, changing only whether $D$ is enabled; each arm spans 8.333 simulated hours. The aggregate target level is anchored to the centre of the policy band; relative prices within each hardware class determine the cross-sectional allocation. The confidence intervals are across the 64 seeded runs. The frozen summary is `data/balancer_ab_64.json`, generated by `node experiments/balancer_ab.js -summary`. MODELLED

Two engineering details matter for whether this is honest.

**The comparison set is the hardware class, not the fleet.** A buyer choosing between accelerators does not consider a general-purpose CPU node "cheap" — they are not substitutes. Comparing against a fleet-wide median would make every CPU look like a permanent bargain and every scarce node look permanently extortionate, and the loop would never settle.

**The curve must be monotone.** If there is any utilisation at which a node gets *cheaper* by becoming busier, a buyer can arbitrage the discount by loading the machine they are about to rent. An earlier version of this design had exactly that defect at one of the seams between the discount, band and premium regions. It was caught by a monotonicity test, and the test is in the repository.




The premium region is superlinear by policy. Its current wear surcharge uses utilisation, an exponential moving average and a hardware sensitivity coefficient. MODELLED It is not a calibrated damage law; production pricing would require temperature and reliability evidence before describing the premium as a maintenance reserve caused by a specific buyer.

### 3.4 $H$ — health

$H$ combines a modeled performance derate, Weibull hazard and correctable-error score. MODELLED The terms are displayed separately in the implementation, but their coefficients are not calibrated against production failures and can overlap with maintenance reserve and capital recovery.

**IN PLAIN ENGLISH: What a correctable error is, and why it predicts failure**

ECC memory can detect and correct specified error patterns. Large-scale field data show that correctable errors are associated with elevated future error risk, but not that every rising count predicts imminent machine failure [4]. Holotrade treats the count as a candidate feature; a production model must control for DIMM age, workload and vendor.

### 3.5 $Q$ — quantum

**IN PLAIN ENGLISH**

This term is 1 by policy for workloads declared classical. For a workload that declares a nonzero non-Clifford budget, $Q$ is a model-specific capability and price field. It is not a universal measure of quantum advantage.

Stabilizer circuits built from Clifford operations admit polynomial-time classical simulation; polynomial does not automatically mean inexpensive for every problem size [3]. PUBLISHED Holotrade's classical routing and accounting code does not require a quantum device. That software fact is separate from any claim that a future error-corrected quantum workload is easy.

The prototype assigns a declared non-Clifford count $t$ the illustrative cost index $9^t$. MODELLED Thus the index is 531,441 at $t = 6$ and about $1.2 \times 10^{19}$ at $t = 20$. This is neither a lower bound for arbitrary circuits nor proof that classical exact or approximate simulation is impossible; cost depends on circuit structure, qubit count, error tolerance and algorithm.

**Why this matters even if you never buy quantum capacity**

It gives the prototype an explicit capability declaration. Classical machines run the exchange's ordinary routing and accounting layer. A future venue could quote separately for a verified quantum service, but no such substrate is provided here.

The prototype refuses to place a plan whose declared capability exceeds a node's declared capability. That is a policy invariant, not a physical impossibility theorem. The refusal code is implemented in software and in the Boolean admission primitive described in §7.

### 3.6 $L$ — locality

$L$ prices the prototype's address-distance from a declared data anchor. Inside one complete cell, graph distance is exact. Across recursive address levels it is a modeled metric, not measured network latency. §4 separates the two.

### 3.7 The floor: what the exchange refuses to do

A two-sided pricing mechanism with an aggressive discount needs a floor, or the balancer will happily discount a machine to a price that never repays it.

$$\boxed{\textbf{floor} = \textbf{electricity} + \textbf{maintenance reserve} + \textbf{capital recovery}}$$

In the prototype, electricity is computed from modeled draw, interval price and PUE; maintenance reserve and capital recovery are policy functions. MODELLED Production use would need accounting treatment, residual value, utilization, service history and safeguards against double-counting the health discount.

**IN PLAIN ENGLISH: What a maintenance reserve is**

Money withheld from current revenue for expected future service. The prototype maps its modeled hazard to a reserve; it has not estimated a causal maintenance cost attributable to a particular buyer.

The prototype **will not post an ask below the floor**, and a regression test asserts that quote invariant. This is an ask-price policy, not evidence of a market-clearing equilibrium.

# 4 The network is the computer

This section separates an exact 40-vertex graph from the prototype's proposed physical and recursive interpretations.

## 4.1 The identity

Holotrade's cell addresses use $W(3,3)$, the symplectic generalized quadrangle over the four-dimensional vector space over $\mathbb{F}_3$. Its collinearity graph has 40 vertices and 240 undirected edges [9].

**IN PLAIN ENGLISH: What that structure is, without the jargon**

Start with the 80 nonzero four-digit vectors over $\mathbb{F}_3$. Identify a vector with its nonzero scalar multiple, leaving 40 projective points; the implementation chooses the representative whose first nonzero coordinate is 1. Two distinct points are adjacent when the alternating form below vanishes.

The exact graph properties used by the prototype are PROVED and exhaustively checked in the repository:

| Property | Value | Why a buyer cares |
|---|---|---|
| Degree | 12 | Each vertex has twelve graph neighbours. A physical design would have to provision or emulate those links. |
| Diameter | 2 | Every pair is at graph distance at most two in the intact, unweighted graph. This says nothing about congestion, queueing or failed links. |
| Non-adjacent paths | 4 | Every non-adjacent pair has four distinct common neighbours, hence four internally vertex-disjoint two-edge paths. |
| Minimum bisection | 100 of 240 | Every 20\|20 cut crosses at least 100 edges, and an explicit cut attains 100. Link rate and traffic determine bandwidth. |
| Adjacency table | none | Adjacency is one arithmetic predicate. Choosing among relays, responding to failures and balancing traffic are separate routing tasks. |

**Address decides adjacency, not the whole route**

For canonical point representatives $u$ and $v$, the predicate is

$$\langle u,v\rangle \;=\; u_0v_1 - u_1v_0 + u_2v_3 - u_3v_2 \pmod 3$$

and two distinct machines are adjacent exactly when it is zero. For a non-adjacent destination, software still selects one of four relays; a deployed network would also need failure, congestion and policy handling.

## 4.2 The consequence: you buy shapes, not counts

Compute throughput and link bandwidth remain different physical resources, but a basket quote can expose their topology jointly. The prototype therefore sells a modeled *shape adjustment* in addition to a node count.

| | **40 scattered machines** | **40 machines as one cell** |
|---|---|---|
| What it is | 40 capacity positions with an arbitrary owned subgraph | The complete 40-vertex cell |
| Worst-case graph distance | may be disconnected | **2** |
| Minimum balanced cut | depends on the owned graph | **exactly 100** |
| Two-edge paths | depends on placement | **4** for non-adjacent pairs |

The same machine count can therefore expose different topology. Holotrade's current *normalized induced-edge coherence* compares the number of graph edges induced by a basket with the model's contiguous-basket baseline. It is a basket property, never a node property.

The current basket-price multiplier is superlinear by design. MODELLED A production calibration would require workload traces and measured link value.

**Exact full-cell bisection; modeled partial-basket coherence**

The spectral lower bound gives 100 for a balanced cut of the complete graph. In the repository's canonical point order, the 20-point side

$$\{2, 4, 5, 7, 8, 9, 11, 12, 16, 17, 19, 21, 23, 24, 28, 29, 32, 35, 38, 39\}$$

has exactly 100 crossing edges, so the bound is attained. PROVED For partial baskets, the implementation uses normalized induced-edge coherence, not the minimum bisection of the induced graph. MODELLED

## 4.3 Which topologies are actually optimal, and can they be delivered

The bisection result above is about the whole cell. A scheduler does not reserve the whole cell — it reserves a subset — so the operational question is sharper: for a request of $m$ nodes, what is the best possible shape, does one exist, can it be placed on whatever is free, and what happens to the leftovers.

All four have exact answers. PROVED

**IN PLAIN ENGLISH: What "optimal shape" means here**

Two reservations of the same size can differ enormously in how well connected they are internally. The extremes are worth naming: the *densest* shape of a given size packs in as many internal links as the geometry permits, which is what one tightly coupled job wants; the *most spread* shape is the one every outside node sees equally, which is what failure-independent replicas want. Everything else sits between them.

Writing $\mathbf{1}_T$ for the indicator of a candidate set and $f = \mathbf{1}_T - \frac{m}{40}\mathbf{1}$, the spectral bounds are attained exactly when $f$ lies wholly in one eigenspace. Using the strongly regular identity $A^2 = 8I - 2A + 4J$, that turns into a pure counting condition on neighbourhoods, and the condition is decidable by exhaustive search:

| | **condition on every node** | **forces** |
|---|---|---|
| densest ($f \in E_2$) | $\lvert N(v) \cap T\rvert = 2 + m/4$ inside, $m/4$ outside | $4 \mid m$ |
| most spread ($f \in E_{-4}$) | $\lvert N(v) \cap T\rvert = 2m/5 - 4$ inside, $2m/5$ outside | $5 \mid m$ |

| $m$ | $\max e(T)$ | $b(T)$ | # shapes | $\tau$ | **busy tolerated** | tilings |
|---|---|---|---|---|---|---|
| 4 | 6 | 36 | 40 | 11 | 10 | 36 |
| 8 | 16 | 64 | 585 | 8 | 7 | 57,132 |
| 12 | 30 | 84 | 5,400 | 8 | 7 | — |
| 16 | 48 | 96 | 21,330 | 6 | 5 | — |
| 20 | 70 | 100 | 33,264 | 6 | 5 | 16,632 |
| 24 | 96 | 96 | 21,330 | 4 | 3 | — |
| 28 | 126 | 84 | 5,400 | 4 | 3 | — |
| 32 | 160 | 64 | 585 | 3 | 2 | — |
| 36 | 198 | 36 | 40 | 2 | 1 | — |

Four things in that table are worth reading off.

**A densest shape exists at every multiple of four, and nowhere else.** So whenever a request is a multiple of four there is a provably optimal shape to serve it, and the repository holds a witness for each.

**The shape counts are a palindrome** — 40, 585, 5,400, 21,330, 33,264, 21,330, 5,400, 585, 40. That is forced rather than coincidental: the complement of a densest shape is a densest shape, so the count at $m$ must equal the count at $40 - m$. The search reproduced it without being told, which is a real check on the search.

**The four-node shapes are exactly the 40 lines of the geometry.** Verified three ways, including brute force over all $\binom{40}{4} = 91{,}390$ subsets. The optimal small reservation is not something the design imposed; it is the incidence structure's own object.

**Perfectly spread sets exist only at $m = 20$.** Every other admissible size is impossible, not merely unfound.

### 4.3.1 Delivery: guarantees rather than averages

A catalogue nobody can place is a wall poster. The symmetry group makes shapes movable: an automorphism preserves adjacency, so it preserves internal link count and boundary *exactly*, and a shape transported onto free nodes carries its optimality with it as a matter of definition rather than approximation.

The column marked **busy tolerated** is the part a scheduler can act on. It is $\tau - 1$, where $\tau$ is the smallest number of blocked nodes that defeats *every* placement of the shape. Below it a placement is certain regardless of *which* nodes are busy; at it, an adversary can block the shape entirely. This is a worst case, not an average.

**Why the four-node guarantee is exactly ten**

The densest four-node shapes are the 40 lines, so defeating them means hitting every line. Each point lies on 4 lines, so $k$ points cover at most $4k$ incidences, and covering all 40 needs $k \geq 10$. *Equality would require all 40 incidences to be distinct* — no two of the ten points collinear — which is precisely an **ovoid**.

This quadrangle has none: its independence number is 7, strictly below the Hoffman ratio bound of 10, established here by exhaustive search rather than by citation. Hence $\tau \geq 11$, and $\tau = 11$ is achieved.

So a four-node reservation **survives any ten simultaneous busy or failed nodes**, and eleven specific ones can defeat it. The bound is tight both ways, and an abstract non-existence result is what fixes an operational number.

The same non-existence caps failure-independent placement: **no eight nodes share zero links**, at any price, on any fabric of this shape. Seven is the ceiling on replica spreading.

### 4.3.2 Fragmentation: the sizes that leave nothing stranded

A tiling is the fragmentation-free schedule — every node used, every reservation provably densest, nothing left between jobs. It needs $4 \mid m$ so a densest shape exists, and $m \mid 40$ so the pieces exhaust the cell, which admits only $m \in \{4, 8, 20, 40\}$ before any search runs.

All three non-trivial cases tile, and the counts are exact: 36 at $m = 4$, 57,132 at $m = 8$, and 16,632 at $m = 20$. The first are the classical *spreads* of the quadrangle, ten disjoint lines meeting every point-star exactly once. The third is exactly half the number of 20-shapes, because each tiles with its own complement — the palindrome above, showing up again as a factor of two.

**An admission policy, not an inequality**

Reserve in units of 4, 8 or 20 and the cell partitions perfectly, with zero stranded capacity and every piece provably densest. At any other size some capacity is stranded by arithmetic: 4 nodes at $m = 12$, 8 at $m = 16$, and more above. PROVED

### 4.3.3 Elastic line ladders: complete on $W(3,3)$, partial on the unitary residues

A spread can do more than tile one static request. Order its ten disjoint lines $L_1, \ldots, L_{10}$ and reserve the prefix $T_i = L_1 \cup \cdots \cup L_i$. The quadrangle axiom makes every pair of lines contribute a perfect matching, so

$$|T_i| = 4i, \qquad k(T_i) = i + 2, \qquad e(T_i) = 2i(i+2), \qquad b(T_i) = 4i(10-i).$$

The boundary is exactly the spectral minimum. Each resize adds or removes one four-point clique, retains the old prefix literally, and therefore migrates no retained point. At $i = 10$ the ladder closes the whole 40-point carrier. PROVED

The same construction applies to $H(3,q^2) = GQ(q^2,q)$, the unitary branch of the $\mathrm{E}_8$ residue programme. For a prefix of $i$ mutually disjoint lines,

$$m = (q^2+1)i, \quad k(T_i) = q^2 + i - 1, \quad e(T_i) = \frac{(q^2+1)i(q^2+i-1)}{2},$$

$$b(T_i) = (q^2+1)i(q^3+1-i),$$

again with equality in the one-sided spectral boundary bound. A GAP 4.12.1 witness reconstructs the two incidence geometries, checks their strongly regular point graphs, exhausts their line-disjointness graphs at the claimed maximum and one rung beyond it, and then verifies every prefix.

| carrier | max. lines | max. points | holes | # maxima | full spread |
|---|---|---|---|---|---|
| $H(3,4) = GQ(4,2)$ | 6 | 30 of 45 | 15 | 72 | needs 9 |
| $H(3,9) = GQ(9,3)$ | 16 | 160 of 280 | 120 | 2,268 | needs 28 |

These are provably partial ladders: there is no seven-line set in the first carrier and no seventeen-line set in the second. The scheduler therefore makes the 15- and 120-point hole sectors explicit instead of presenting the last rung as full coverage. Its topology-plan envelope is tamper-evident and non-dispatchable until the abstract GAP points are bound to inventory and the resulting physical topology is independently attested. PROVED

The ceiling leaves structure, not merely unused points. In each carrier the ambient automorphism group is transitive on the maximum partial spreads, so the hole graph has one isomorphism type. For $q = 2$ it is $\mathrm{SRG}(15,6,1,3) \cong KG(6,2)$, with automorphism group $S_6$. For $q = 3$ it is a 20-regular diameter-two graph on 120 vertices, with spectrum

$$20^1,\ 8^5,\ 4^{45},\ 0^9,\ (-4)^{60}.$$

It is not strongly regular. Its full automorphism group is

$$G \cong (\mathbb{F}_2^6/\langle \mathbf{1} \rangle) \rtimes S_6 = \mathrm{Aut}(\text{folded } Q_6), \qquad |G| = 23{,}040,$$

and GAP identifies the graph exactly as the coset action $G/H$, where $H =$ `SmallGroup(192,1485)` and adjacency is the union of the degree-16 and degree-4 orbitals. The maximum-spread stabilizer has order 11,520; it is not the full hole-graph group. PROVED

The most obvious 120-state folded-cube model does not work. The point-duads, equivalently antipodal square faces, have stabilizer `SmallGroup(192,1472)` $\cong S_4 \times D_8$, and their unique degree-20 two-orbital graph is not isomorphic to the hole graph. This executable no-go isolates the correct permutation representation: the symmetry bridge is exact, but the natural-duad carrier is wrong. It is not a claim about a K3 surface or a deployed folded-cube network. PROVED

#### 4.3.4 The cross-prime fibre theorem and a 40–40–40 differential

The two hole carriers are not merely analogous. Let $N = O_2(G) \cong C_2^5$ in the full $q = 3$ hole group $G = 2^5{:}S_6$, and let $H$ be the order-192 coset stabilizer above. Exact GAP calculation gives

$$|H \cap N| = 4, \qquad N/(H \cap N) \cong C_2^3, \qquad HN/N \cong C_2 \times S_4,$$

so the $N$-orbits form fifteen fibres of eight states and the induced block action is $G/N \cong S_6$. Every fibre induces $K_{4,4}$. Between two fibres whose duads meet, the *cross-only* bipartite connector is $4C_4$; disjoint-duad fibres have no connecting edge. Including both internal $K_{4,4}$ graphs makes the induced two-fibre graph connected and 6-regular. The former “$C_{16}$” description confused its 16-vertex connected union with the cross-only graph type and is retracted here. PROVED

More precisely, if $Q_{ij}$ is the number of neighbours in fibre $j$ of any one vertex in fibre $i$, then the partition is equitable and

$$Q = 4I + 2A(T(6)).$$

The off-diagonal zero relation is therefore the complement of $T(6)$, namely $KG(6,2)$ — exactly the $q = 2$ hole graph. This also explains the quotient spectrum

$$20^1,\ 8^5,\ 0^9 :$$

the nine-dimensional rational zero eigenspace is the block-constant lift of the eigenvalue-1 space of the $q = 2$ Kneser graph. This is an object-level cross-prime descent, not a shared-order observation. PROVED

The same adjacency matrix has a different exact life in characteristic two. For its reduction $A_2$ over $\mathbb{F}_2$, GAP obtains

$$A_2^2 = 0, \qquad \operatorname{rank} A_2 = 40, \qquad \dim\ker A_2 = 80.$$

Consequently

$$0 \subset \operatorname{im} A_2 \subset \ker A_2 \subset \mathbb{F}_2^{120}$$

has associated graded dimensions $40 \mid 40 \mid 40$, and the middle homology $\ker A_2/\operatorname{im} A_2$ is 40-dimensional. The scheduler exposes this as an exact parity switch: one application lands in the image and the second vanishes. It remains an abstract logical transform, not a reversible gate or a command to a host. In particular, the dimension match does *not* identify any graded piece with the forty points of $W(3,3)$; that would require an explicit equivariant map. PROVED

The fibre partition itself is no longer a coordinate choice. GAP reconstructs the full graph automorphism group, takes its characteristic subgroup $O_2(G) \cong C_2^5$, and recovers exactly the same fifteen orbits of size eight. Thus the unlabelled $15 \times 8$ partition is intrinsic to the graph. Only the labels of the fifteen blocks and the eight slots inside each block remain gauge choices; canonical blocks do not bind those logical states to machines. PROVED

### 4.3.5 Affine relation compiler and reversible dilation

The block quotient can be compiled more tightly than the general $GF(9)$ predicate. Each of the sixty edges of $T(6)$ carries a codimension-two affine relation on $\mathbb{F}_2^3 \times \mathbb{F}_2^3$. Each relation contains eight affine permutation graphs and has four complementary two-map decompositions; choosing one decomposition gives the two parallel channels whose union is $4C_4$. Across all base edges only thirteen relation types occur. This is an affine multichannel relation lift, not a regular one-voltage cover in the standard sense [11]. PROVED

The generated RTL evaluates the thirteen relation types in 171 iCE40 LUT4s and four carry cells under the recorded no-ABC flow. The older general coordinate predicate uses 720 LUT4s under that same flow, a reduction of 76.25%. A Yosys SAT miter proves the affine predicate equivalent to an independent complete 120-row reference over all $2^{14}$ address pairs. These are logic counts and exhaustive Boolean equivalence, not timing, power or board measurements. PROVED

Although $A_2$ is square-zero rather than invertible, it has a canonical reversible enlargement on 240 bits:

$$U(x, y) = (x + A_2 y, y), \qquad U = \begin{pmatrix} I & A_2 \\ 0 & I \end{pmatrix}.$$

Symmetry and $A_2^2 = 0$ give $U^2 = I$ and $U^\mathsf{T}\left(\begin{smallmatrix} 0 & I \\ I & 0 \end{smallmatrix}\right)U = \left(\begin{smallmatrix} 0 & I \\ I & 0 \end{smallmatrix}\right)$. GAP verifies $\operatorname{rank}(U - I) = 40$ and fixed-space dimension 200. The 2,400 CNOTs form a 20-regular bipartite graph, which the witness decomposes into twenty perfect matchings of 120 gates. Depth 20 is optimal because every wire participates in twenty gates; Yosys independently proves that two RTL applications are the identity for every 240-bit input. This is an exact logical reversible circuit, not a quantum-hardware performance claim. PROVED

### 4.3.6 Control codec, orientation boundary and compiled graph

The seven-point Fano control plane admits the exact factorization

$$\mathrm{GL}(3, 2) = (C_7{:}C_3)D_8, \qquad (C_7{:}C_3) \cap D_8 = 1.$$

Neither factor is normal. Thus its 168 states have one $21 \times 8$ address chart, but multiplication is the nontrivial matched-action, or Zappa–Szép, product rather than componentwise multiplication. The GAP witness checks all 28,224 products and all 4,741,632 associativity triples before JavaScript freezes the table. These are abstract control-plane addresses and cannot name machines without a separately signed inventory binding. PROVED

The index-two extension on the 120-state hole carrier also needs careful wording. The embedded maximum-spread stabilizer $2^4{:}S_6$ is normal in the full $2^5{:}S_6$ action and supplies a relative $C_2$ orientation character. The 2,268 maximum spreads group into 1,134 unordered fibres: the two members cover the same points, share no lines and are exchanged by an equivariant partner involution. But the unmarked full hole group has abelianization $C_2 \times C_2$ and three index-two normal subgroups, so it supplies no intrinsic zero/one label. Nor does it select a rollback: 376 outer involutions split into five conjugacy classes under the spread subgroup and the centralizer of the hole action is trivial. Runtime rollback must therefore name and attest an explicit class and representative. PROVED

Finally, the exact 120-state graph now has two independent synthesizable implementations. The compact circuit stores 120 normalized $GF(9)^4$ coordinates (1,920 bits) and evaluates Hermitian orthogonality; the reference circuit stores all 120 adjacency rows (14,400 bits). A Yosys SAT miter proves equality over every one of the $2^{14}$ input address pairs. Current iCE40 synthesis reports 544 LUT4 cells for the compact predicate and 1,174 for the row ROM. These are logic counts, not timing, power, board or physical-network measurements. PROVED

#### 4.3.7 The full normalizer closes abstractly, not at chain level

The companion normalizer calculation is also exposed fail-closed. On the characteristic-two dihedral branch $D_{26} = C_{13}{:}C_2$, the complete extension defect lies in the 316-dimensional $C_{13}$-fixed sector:

$$F_2[V_2]|_{C_{13}\text{-fixed}} = J_1^{64} \oplus J_2^{126}, \qquad H_1|_{C_{13}\text{-fixed}} = J_2^{158}.$$

The complementary $W_{12}^{315}$ sector already agrees, yielding the exact stable identity

$$F_2[V_2] \oplus J_2^{32} \cong H_1 \oplus J_1^{64} \quad \text{as } F_2[D_{26}]\text{-modules,}$$

at dimension 4,160 on each side. Both correction carriers are external with trivial $C_{13}$ action. The natural $\mathbb{F}_4^3$ fixed cone cannot replace them: three locally $C_3$-compatible translation pairings exist, but no nonzero translation is $C_{13}$-equivariant. PROVED

The restrictions nevertheless assemble at the level of abstract modules. For the full semidirect product $C_{13}{:}C_6$, with the complement acting on $C_{13}$ by $a \mapsto a^4$, GAP decomposes both 4,096-dimensional carriers into the four indecomposables obtained from $J_1, J_2$ and the nontrivial two-dimensional $C_3$ module. A 64-dimensional correction on each side gives one exact stable isomorphism at dimension 4,160. Its $C_2$ restriction recovers $J_2^{32}$ versus $J_1^{64}$; the former consists of 32 copies of the unique nonsplit class in $\mathrm{Ext}^1(\mathbf{1},\mathbf{1})$, while the Tate dimension on the latter side is 64, so the numerical defect is explained by $64 = 2 \cdot 32$. Its $C_3$ restriction recovers the independently computed odd branch. PROVED

This closes the abstract $\mathbb{F}_2[C_{13}{:}C_6]$ stable-module question. It does *not* give a direct 4,096-dimensional bridge, a 4,160-by-4,160 chain map, or a dispatchable scheduler transform. A tempting 32-cycle Hall–Janko carrier is also rejected: its $C_3$ cycle profile differs from the local extension-pair profile. The runtime therefore returns the stable plan and names an explicit chain-level intertwiner as the next required evidence.

#### 4.3.8 A direct-product recursive control chart

The 168-state Fano matched-action controller and the 120-state execution switch compose into the bijective chart

$$21 \times 8 \times 15 \times 8 = 20{,}160.$$

This is not merely multiplication of state counts. The first action group is the simple group $\mathrm{PSL}(3,2)$ of order 168; the second has order 23,040 and has no quotient of order divisible by 7. They therefore have no common nontrivial quotient, and Goursat's lemma forces every subdirect coupling that surjects onto both factors to be their direct product. The resulting product action has order 3,870,720 and point stabilizer 192. The coincidence $20{,}160 = |A_8|$ is not an identification: the states are Cartesian logical addresses, not $A_8$ elements, hosts or microVMs. PROVED

Adding one native 40-state $W(3,3)$ router digit gives an 806,400-state local logical carrier. A second Goursat calculation proves that connected $\mathrm{PSp}(4,3)$ routing is forced to be independent of the engine action. For the full order-51,840 router group there are exactly four subdirect classes: the direct product and three index-two fibre products, selected respectively by permutation sign, translation parity and their XOR. All three parity pullbacks are noncanonical; no router vertex or engine state selects one, so the result supplies neither a recursive parity policy nor dispatch authority. PROVED

#### 4.3.9 One 85-state polarity plane and its binary sentinel

The two carrier sizes $40 + 45$ are blocks of one exact matrix, not merely a numerical partition. Let $B$ be the 40-by-45 incidence matrix whose columns are the eight-point trade supports. Direct GAP reconstruction from $W(3,3)$ gives

$$BB^{\mathsf{T}} = 8I_{40} + 2A_{W(3,3)} + J_{40}, \qquad B^{\mathsf{T}}B = 8I_{45} + 2A_{\overline{GQ(4,2)}}, \qquad \mathrm{rank}_{\mathbb{Q}}\, B = 25.$$

Thus the loopless chiral operator with off-diagonal block $B$ has kernel split $15 + 20$ and spectrum $0^{35}$, $\pm\sqrt{72}^{1}$, $\pm\sqrt{12}^{24}$. PROVED

Only five of those 35 zero modes are protected by the rectangular chiral index under an arbitrary off-diagonal perturbation: every 40-by-45 matrix has nullity at least $45 - 40 = 5$. The witness adds the 40-by-45 diagonal embedding to $B$ and obtains rank 40, leaving exactly five zero modes in the doubled chiral operator. Thus the other thirty zeros require the special incidence rank/module symmetry. This is a finite perturbation theorem, not evidence for a physical flat band. PROVED

Restoring the polarity loops on the 45 absolute points completes the block matrix to an 85-by-85 point–polar-plane incidence matrix $H$. The same witness checks entry by entry

$$H^2 = 16I_{85} + 5J_{85}.$$

Hence $H$ is the symmetric $2-(85,21,5)$ design, with spectrum $21^1, 4^{45}, (-4)^{39}$, determinant $-21\,4^{84}$ and $H^{-1} = H/16 - 5J/336$. PROVED

Over $\mathbb{F}_2$, the columns of $B$ span a doubly-even self-orthogonal $[40,15,8]_2$ code. Enumerating all $2^{15}$ words gives

$$1 + 45z^8 + 720z^{12} + 6930z^{16} + 17376z^{20} + 6930z^{24} + 720z^{28} + 45z^{32} + z^{40},$$

and set comparison proves that the 45 columns of $B$ are exactly all 45 minimum words. Their Hamming metric reconstructs $GQ(4,2)$: distance 16 means disjoint support, while distance 12 means intersection size two. The resulting rank-15 binary matroid has girth five and exactly 216 five-circuits; each is a five-coclique, every nonedge lies in three circuits, and GAP verifies the orbit $\mathrm{PSp}(4,3)/S_5$. PROVED

The projective design and invariant code are classical [12]; the certified integration result is their explicit equality in the same $B$ coordinates already used by the W33 and $GQ(4,2)$ schedulers. The 216-element circuit orbit is not identified with the order-216 projective qutrit Clifford group, and its parity plans are not cryptographic authentication.

#### 4.3.10 The 120-state homology, its sentinel shadow and the two 216s

For the same 120-state adjacency differential over $\mathbb{F}_2$, exhaustive enumeration and the binary MacWilliams transform give

$$\operatorname{im} A = [120, 40, 16]_2, \qquad \ker A = [120, 80, 8]_2, \qquad \ker A / \operatorname{im} A \cong \mathbb{F}_2^{40}.$$

Taking the same basis of $\operatorname{im} A$ for both CSS checks gives the exact code $[[120, 40, 8]]_2$: all 435 weight-eight kernel words are nontrivial logical classes because the image has minimum weight 16. This is a classical stabilizer-code theorem; no decoder or noise model is supplied. PROVED

The equality of dimensions does not identify this homology with the 40-point $W(3,3)$ permutation module. GAP tests both inner and exceptional-outer $S_6$ identifications and rejects an isomorphism. Instead, every equivariant map from the homology to either genuine point carrier kills the $C_2^5$ translation relations, factors through a 15-dimensional coinvariant quotient, and lands inside the explicit $[40, 15, 8]_2$ sentinel code. The span of all forward images has dimension 11 in the inner class and 6 in the outer class; every reverse map lands in the 15-dimensional fixed homology, with the same two universal image dimensions. No preferred nonzero map is selected. PROVED

The bicolour maximal-overlap matrices $M_+$ and $M_-$ on the 216 five-circuits have exact rational ranks

$$\operatorname{rank} M_+ = \operatorname{rank} M_- = 216, \qquad \operatorname{rank} \begin{bmatrix} M_+ \\ M_- \end{bmatrix} = 372.$$

Thus their common row space has dimension 60 and module shape $1+15+20+24$. Reconstructing the established 216-by-45 circuit incidence inside the same GAP witness proves more:

$$\operatorname{im}(C)M_+ = \operatorname{im}(C)M_-, \quad \dim = 45, \quad \operatorname{im}(C) = 1 + 20 + 24.$$

Consequently the common 60-space is this colour-independent circuit 45-space direct-summed with the diagonal 15. This also repairs the earlier use of one finite-field rank as if it were a rational equality. PROVED

There is an equally important separation. The 432 $W(3,3)$ hemisystems give 216 projective lines after quotienting by complementation, but these are *not* the 216 circuit states. Both actions are transitive and both $\mathrm{PSp}(4,3)$ stabilizers are abstractly $S_5$ of order 120, yet the stabilizers are nonconjugate. In the full graph automorphism group they grow to order 240 and remain nonconjugate; eleven conjugacy classes have different fixed-point counts. In particular, one involution class fixes 48 circuits and no hemisystem lines, while another fixes 6 circuits and 66 hemisystem lines. Equal cardinality therefore does not define an equivariant dictionary, even after the outer automorphism is allowed. The irreducible decomposition makes the separator canonical: the circuit permutation module contains the 81-dimensional Steinberg representation once, while the hemisystem-line module contains it zero times. This is the same Steinberg species isolated independently in the 1,080-state obstruction-product carrier of the companion W33 computation. PROVED

The common quotient is now constructed rather than inferred from the equality $216 = 36 \cdot 6$. Both carriers map equivariantly onto the same 36 regular spreads (equivalently, the 36 Schlaefli double-sixes); every fibre has six states and its spread stabilizer is $S_6$. The two order-120 point stabilizers are the two nonconjugate $S_5$ classes inside $S_6$. The exceptional outer automorphism of the abstract fibre group exchanges those classes, but exhaustive conjugation shows that no automorphism of the $W(3,3)$ substrate realizes that exchange. The two 216-state machines therefore form a genuine carrier fork, not two gauges of one carrier. PROVED

There is nevertheless a canonical bridge: their fibre product over the 36-state base,

$$G/S_5^{\mathrm{circ}} \times_{G/S_6} G/S_5^{\mathrm{pair}} \cong G/(S_5^{\mathrm{circ}} \cap S_5^{\mathrm{pair}}), \qquad |G/F_{20}| = 1,296,$$

where GAP identifies the intersection as $F_{20} \cong C_5{:}C_4$. Frobenius reciprocity gives a sharper cross-repository synthesis. If $H_{81} = H_1(W(3,3))$ and $H_{64} = H_1(GQ(4,2))$, then the 1,296-state permutation carrier contains

$$3H_{81} \oplus 3H_{64}, \qquad 3 \cdot 81 + 3 \cdot 64 = 435.$$

The corrected 1,080-state obstruction carrier contains exactly the same building block, so these 435-dimensional isotypic components are abstractly $G$-isomorphic and their equivariant cross-Hom space has dimension $3^2 + 3^2 = 18$. This computation also catches a subtle coordinate error in the first HoloTrade witness: the certified obstruction is 27 completion charts times 40 *lines*, not 40 points. The point product has $H_{64}$-multiplicity four and is a different $G$-set. An explicit $1,296$-by-$1,080$ intertwiner is not claimed here. PROVED

The companion W33 repository already supplies a typed software ABI over the 36-state base and same-carrier versus neutral-continuation checkpoint semantics. The additional result here is the exact group-action hardware lift. The corresponding finite controller is executable on both forks. Two native generators act on each 216-state carrier, while a cross-carrier request returns only the common 36-state quotient and never fabricates a forbidden 216-to-216 conversion. Yosys proves the quotient-commutation, range, and fail-closed adapter assertions for all carrier, opcode, state and adapter inputs; a control that exposes an invented cross-carrier state produces a SAT counterexample. This is a combinational logic theorem, not a processor, benchmark, or fabricated-device claim. FORMAL

### 4.3.11 The fibre stabilizer is a protected five-qutrit control group

The order-20 stabilizer above is not merely a label on the scheduler fibre. The cyclic $[[5,1,3]]_3$ block is the qutrit instance of the nonbinary five-letter single-error-correcting family [13]. GAP exhaustion gives its bare coordinate-permutation image as only $D_{10}$. Allowing an independent local $SL(2,3)$ Clifford on each site enlarges the coordinate image to

$$F_{20} = C_5{:}C_4.$$

More strongly, explicit generators $T, M$ on the ten-dimensional qutrit Pauli space preserve both the stabilizer row space and the symplectic form and obey

$$T^5 = M^4 = 1, \qquad MTM^{-1} = T^3.$$

These are the same presentation and exponent as the fibre-product point stabilizer, so $a \mapsto T$, $b \mapsto M$ defines a generator-level isomorphism rather than an order match. The missing order-four affine multiplier is invisible to bare site permutations and is supplied exactly by the local Clifford compensation plus an explicit stabilizer-row change.

The faithful action on five addressed sites times the eight nonidentity single-qutrit Paulis is compiled into a 40-state controller. Yosys proves closure and all three presentation relations for every valid address. A mutation that replaces the compensated multiplier by the identity produces a counterexample. Thus a fibre state has a native protected block-control action. This does not identify the $1,296$ fibre states with codewords, make the nonlocal 20-to-240 qutrit embedding local, or close calibration, threshold, or fault-tolerant recoding. PROVED FORMAL

#### 4.3.12 One $F_{20}$, two five-state geometries

The Payne-derived slow-path certificate makes the protected-block statement object-level. On the 45 expensive targets, the inner fibre $F_{20}$ has orbit lengths

$$5 + 10 + 10 + 20.$$

The five-orbit is a coclique of $GQ(4,2)$, its set stabilizer induces the full $S_5$, and its orbit under $PSp(4,3)$ contains 216 sets. It is therefore one of the already constructed five-target circuit states rather than one of the 27 five-point ROM lines. With the generator identification above fixed, GAP finds exactly one equivariant bijection from the five qutrit sites to this circuit.

A ROM line supplies a second, inequivalent five-state action. Its stabilizer has order 960 in $PSp(4,3)$, with elementary-abelian kernel $2^4$ and image $A_5$ on the line. In the full order-51,840 Schlaefli/Weyl action the stabilizer has order 1,920 with the same kernel and image $S_5$. The preimage of $AGL(1,5) < S_5$ has one conjugacy class of complements $F_{20}$, and the generator-level calculation gives

$$F_{20} \cap PSp(4,3) = D_{10}.$$

Under the explicit site-to-line map this is the same subgroup pair as the bare $D_{10}$ and locally Clifford-completed $F_{20}$ of the qutrit block. The parallel is algebraic: it does not identify a local Clifford operation with a physical $W(E_6)$ transformation. PROVED

There is also an exact forty-address transport. The nonidentity one-site Paulis and the W33 axes each split into two regular $F_{20}$-orbits, giving exactly 800 equivariant bijections. A frozen lexicographic choice maps each Pauli address to a W33 axis and hence to its nine-target Payne cover. This choice is reproducible, not geometrically canonical. Exhaustion over all 20 presentation-compatible $F_{20}$ identifications and all 16,000 resulting map candidates proves a useful no-go: none simultaneously sends the eight Paulis above every site onto the eight Payne covers containing that site's matched circuit target. Thus the site-to-circuit and address-to-axis maps both exist, but $F_{20}$ symmetry alone does not weld them into the stronger incidence codec. This proper-subgroup construction is consistent with the companion repository's full-$PSp(4,3)$ characteristic-zero 27-to-40 linear no-go; it does not evade that theorem.

The exhaustive defect spectrum is unexpectedly rigid. Of the 16,000 candidates, 9,600 match none of the forty address incidences. The other 6,400 match exactly four of eight addresses at *each* of the five sites, hence twenty overall. No intermediate profile occurs. The optimum is therefore a uniform half-incidence map, not an almost-correct map concentrated on four sites. For the frozen zero-match gauge, composing with the independently proved Payne staging rule moves every address axis to an axis covering its matched circuit target in exactly one W33-collinearity hop. That composition is a certified operational repair, but not a direct $F_{20}$-equivariant weld, and it remains non-dispatchable. PROVED

#### 4.3.13 The all-transvection cost anomaly is a nondegenerate reflection

The current finite instruction metric is the Cayley metric generated by all 80 nontrivial transvections of $Sp(4,3)$, hence by both scalar conjugacy classes. Exhaustive enumeration of all $51,840$ elements proves

$$\ell(g) > \operatorname{rank}(g-I) \quad\Longleftrightarrow\quad g|_U = -I, \quad g|_{U^\perp} = I$$

for a nonzero nondegenerate subspace $U$. In dimension four, $U$ has dimension two or four. The first case supplies one distinguished reflection on each of the 90 hyperbolic lines, and the second supplies the central $-I$, accounting for all 91 anomalies by one rule. A hyperbolic image is necessary but not sufficient: each hyperbolic line carries fifteen residue-two elements with length profile 14 of length two and one of length three. This corrects the earlier class-level dichotomy. The rule fails at $q = 2$, where $-1 = +1$.

Ellers determined the exact $\lambda$-length of every element of $Sp(2n,3)$ when the generating transvections are restricted to one fixed conjugacy class [14]. That theorem is direct prior art. The two-class 80-generator metric used here is different; the contribution of the exhaustive computation is its exact anomaly set and the explicit map $g \mapsto \operatorname{im}(g-I)$ to the nondegenerate-line geometry, not a claim that transvection-length classification was absent from the literature.

The extremal bound $(q^3+q+2)/2$ and its sharpness for $q = 2,3$ are published finite-geometry results, not a claim of this paper [10]. The contribution here is the executable scheduler lift, the exact residual graph/group/coset classification, and their evidence boundary. The standard Hermitian models are incidence-isomorphic to the $\mathrm{E}_8$ residue carriers certified in the companion W33 repository; no literal $\mathrm{E}_8$ coordinate isomorphism is constructed in this implementation.

> **Scope of the $W(3,3)$ catalogue above.** These are exact statements about the level-1 40-point graph: reservations of *points*, verified by three independent routes — combinatorial search in JavaScript, eigenspace projection in Python, and GRAPE/nauty in GAP, which also settles that the graph's full automorphism group has order 51,840 and that every shape orbit is unchanged under it, so the guarantees are exact rather than conservative. They say nothing about link-state routing, congestion-aware forwarding, physical recovery, or deployed placement across an address hierarchy. Healthy level-1 recovery controllers and failure-aware policy experiments are built and reported separately; they do not turn the point-graph catalogue into a network implementation.

### 4.4 What happens above one cell, and why it does not scale

Everything above concerns a single 40-node cell. A fleet is not one cell, so the question is what survives when the fabric is built recursively. The answer is sharp, and it is mostly negative. PROVED

**IN PLAIN ENGLISH: The recursive fabric**

> Each of the 40 positions is replaced by a whole copy of the structure, giving $40^n$ leaves at level $n$. Two leaves are adjacent when they sit in the same cell and are adjacent inside it, or when they sit at the same position in adjacent cells. That is the homogeneous wiring — no gateway node, no privileged position.

At level $n$ the fabric is $12n$-regular with second eigenvalue $12n - 10$, so

$$k \;-\; \lambda_2 \;=\; 10 \quad \text{at } \textit{every} \text{ level}$$

The spectral gap never grows. Everything below follows from that one fact.

| $n$ | leaves | degree | $\lambda_2$ | gap | expansion | smallest shape | survivable |
|---|---|---|---|---|---|---|---|
| 1 | 40 | 12 | 2 | 10 | 0.8333 | 4 | 25.00000% |
| 2 | 1,600 | 24 | 14 | 10 | 0.4167 | 160 | 0.62500% |
| 3 | 64,000 | 36 | 26 | 10 | 0.2778 | 6,400 | 0.01562% |
| 4 | 2,560,000 | 48 | 38 | 10 | 0.2083 | 256,000 | 0.00039% |
| 5 | 102,400,000 | 60 | 50 | 10 | 0.1667 | 10,240,000 | 0.00001% |

### 4.4.1 Every optimal shape above level one is a replication

A set attains the densest bound exactly when its indicator, minus a constant, lies wholly in the $\lambda_2$-eigenspace — and for a Cartesian power that eigenspace consists of a level-1 eigenvector placed in *one* coordinate. So the indicator must be a sum of functions of separate coordinates. Since it takes only the values 0 and 1, and $|A_1 + \cdots + A_n| \geq \sum |A_i| - (n-1)$ for finite sets of reals, at most one of those functions can be non-constant.

**The classification, and what it forbids**

Every densest reservation above level one is a level-1 optimal shape replicated along one coordinate. **There is no third kind**, so there is no cleverer shape waiting to be found.

Three consequences, all bad for large fleets. Expansion decays as $10/(12n)$, so the fabric gets structurally worse at connecting itself as it grows. The ladder of available sizes is $40\times$ coarser at each level — at level 5 the smallest provably densest reservation holds **10,240,000** leaves and there is nothing below it. And every one of them tolerates the same **ten** simultaneous failures, which is 0.00001% of a level-5 fabric.

The practical reading is that provably-densest reservations are a **level-1 tool**. Above that, the honest move is to compose level-1 shapes inside cells and accept a sub-optimal joint. The implementation says exactly that: `levelAdvice()` refuses off-ladder requests, reports the survivable fraction, and recommends the alternative rather than quoting a bound nobody can use.

**Prior art.** None of this machinery is new. Intriguing and tight sets are Payne (1987) and Bamberg–Kelly–Law–Penttila; the same objects appear as equitable 2-partitions, where the eigenfunction characterisation used here is standard, and equitable 2-partitions of Cartesian powers have their own literature. The separability step is elementary and likely folklore. What is claimed is not new mathematics but that *this* fabric is now classified, so the architectural question has a definite answer.

**Scope.** These are statements about the level-1 point graph and its Cartesian powers. The repository's software distance model, $16n-14$, measures a different quantity — a declared cost in reversible control moves — and the measured graph diameter at level 2 is 4. Both are fine; conflating them is not.

### 4.5 The mechanism this creates: a defragmentation market

**IN PLAIN ENGLISH**

Trading entitlements can fragment a participant's accessible induced subgraph, even though it does not change the physical links. The demo seeds fragmented portfolios to illustrate this allocation problem. MODELLED

The fix is a **swap book**. I give you my isolated machine sitting in *your* block; you give me yours sitting in *mine*. No cash changes hands beyond a small adjustment for the difference in condition between the two machines. The prototype offers a swap only when both modeled portfolio scores rise. Physical network capacity does not change.

This resembles entitlement defragmentation. Real acceptance would also depend on hardware class, data locality, legal constraints and transaction costs.

**A swap is only offered when both sides gain**

The engine computes both parties' normalized induced-edge score before proposing a swap, and a regression test asserts both scores increase. This is an invariant of the prototype objective, not proof of economic welfare.

### 4.6 The recursive address model

**IN PLAIN ENGLISH**

A level-$n$ software address is an $n$-digit base-40 path. The current distance model charges at most two graph edges at the first divergent digit and eight abstract reversible moves for every remaining suffix digit on both sides. It does not yet specify a physical graph product, switches or links.

The address count $40^n$ is exact. Under the current software metric, equal-depth addresses have the bound

$$d_n \leq 2 + 16(n-1) = 16n - 14.$$

MODELLED This is an address-cost bound, not a physical hop theorem.

| Level | Machines | Worst-case hops | Seats |
|---|---:|---:|---|
| 1 | 40 | 2 | one address cell |
| 2 | 1,600 | 18 | modeled |
| 3 | 64,000 | 34 | modeled |
| 4 | 2,560,000 | 50 | modeled |
| 5 | 102,400,000 | 66 | modeled |
| 6 | 4,096,000,000 | 82 | modeled |
| 7 | 163,840,000,000 | **98** | modeled |

**One prototype interface at every modeled scale**

A software `Composite` exposes the same quote interface as a leaf: an address, modeled history, health, utilisation and throughput. The same pricing function can therefore quote both without type-specific branches.

The repository test quotes a leaf, cell and composite through that one code path. This proves interface closure only; it does not prove that a physical campus is one machine or that liquidity aggregates self-similarly.

Any deployed service-level agreement would still need measured latency, availability, congestion and remedies. The model's address bound is not an SLA.

## 4.7 Joining is local in the prototype

A machine can fill an unused modeled address slot with a local registry update. A complete $W(3,3)$ cell has exactly 40 positions, so a 41st machine requires another cell, a replacement or a different topology. Physical joining would also require cabling, link configuration, identity and failure handling.

Point transitivity means graph positions are structurally equivalent; it does not make heterogeneous machines operationally indistinguishable. The local splice test is therefore an implementation invariant, not a zero-cost claim.

# 5 Instruments

The prototype renders five order or contract templates. Only the first is a spot capacity purchase; the supply offer is seller-side. Forward, option and lease prices below are scenarios, not validated valuation models. They would require delivery, assignment, collateral, cancellation, exercise, default and jurisdiction-specific regulatory terms.

| Instrument | Tenor | The risk it addresses |
|---|---|---|
| **Spot node-second** | immediate | Capacity metered and settled by the second, with an hourly-equivalent display rate and a separate cold-start charge. |
| **Forward block** | 1–90 days | An illustrative reservation for future capacity. Compute is not a freely storable commodity, so the displayed spot-plus-risk rule is a model rather than a no-arbitrage cost-of-carry theorem. |
| **Burst option** | 1–30 days | You *might* need 400 machines on launch day. You want the right, not the obligation. A production premium requires a specified payoff distribution, exercise rule, reservation policy and capacity correlation; the demo uses a simple scenario rule. |
| **Genome lease** | 7–365 days | You want versioned software state with a history on your workload class. Its premium remains a hypothesis pending out-of-sample workload validation. |
| **Supply offer** | open | You own idle machines and post capacity subject to the modeled quote floor. This is an order type, not a buyer-risk instrument. |

**The margin hypothesis**

Competitive undifferentiated capacity may face pressure toward marginal and capital-recovery cost, but convergence is not guaranteed and the prototype is not evidence of market structure.

Reusable software state could support differentiation if it produces verified, portable, out-of-sample improvement after accounting for contamination and customer-data restrictions. That is the main commercial hypothesis to test, not a result established here.

### 5.1 A policy mobility score, not observed liquidity

**IN PLAIN ENGLISH**

Market liquidity depends on counterparties, order flow, prices, substitutability and time. Before trading starts, the rule engine can compute only a narrower quantity: the fraction of address positions permitted by declared policy.

Let $N_{\text{eligible}}$ be the number of the 40 cell positions allowed by a specific prototype policy configuration. The displayed mobility score is $M = N_{\text{eligible}}/40$. MODELLED The eligible counts below are policy scenarios. They are not group orbits, stabilizer quotients or measurements of market liquidity.

| Policy scenario | Eligible | $M$ | Display band |
|---|---:|---:|---|
| none | 40 | 1.000 | deep |
| GPU affinity | 27 | 0.675 | liquid |
| single-tenant | 20 | 0.500 | liquid |
| data residency | 13 | 0.325 | thin |
| air-gapped | 5 | 0.125 | bilateral |

**What the score is useful for**

$M$ is an auditable policy disclosure: it tells a buyer how many modeled positions satisfy the declared constraints. It can flag a potentially narrow delivery set. It must not be labelled liquidity or used as a price discount until calibrated against actual fills, spreads and market depth.

# 6 Energy and the thermodynamic floor

**IN PLAIN ENGLISH**

Compute is bought in dollars and paid for in joules. Data centres drew about 415 terawatt-hours globally in 2024 — roughly 1.5% of world electricity. The International Energy Agency projects about 945 TWh by 2030 under its base case [2]. PUBLISHED

That is a scenario, not a physical law. Holotrade's narrower claim is that a quote can expose the energy price interval, PUE assumption and modeled joules instead of hiding them in one rate.

## 6.1 The honest efficiency denominator

**IN PLAIN ENGLISH: The Landauer limit**

Landauer derived a lower bound $kT \ln 2$ for erasing one bit under the stated thermodynamic assumptions [5]. Bennett showed how logically reversible computation can in principle avoid compulsory erasure at each logic step [6]. Neither result removes loss, control, speed, noise or error-correction costs from a physical device.

The source model assigns 58 ternary records per error-correction cycle and assumes each record is eventually reset, with conditional lower bound $kT \ln 3$ per reset. Measurement alone is not necessarily erasure; record reset is the relevant step. Under that model, at room temperature the arithmetic is

$2.64 \times 10^{-19}$ joules per modeled cycle at 300 K MODELLED

This per-cycle conditional bound cannot be compared directly with a vendor's joules per operation until "cycle", useful operation, clock rate and all control and memory costs are mapped consistently.

**How the denominator may be used**

The Landauer expression is a stable physical reference once the logical erasure event is defined. It is not a workload-efficiency benchmark and can mislead if vendors count different operations or omit memory, cooling and control.

The demo displays a conditional orders-of-magnitude gap for exploration. A production quote should pair it with useful-work-per-joule, total facility energy and the exact operation definition.

**Honest scope.** The number is conditional on the model's 58 reset records. It is not measured device efficiency, a throughput result or evidence that only those erasures occur. No full build of the proposed architecture exists.

# 7 The hardware

The prototype's combinational routing and admission-policy classifier is small enough to synthesize for an FPGA. It consumes predicate results; it does not perform digital-signature verification, artifact hashing, replay-store lookup or remote attestation itself.

The RTL computes the cell adjacency/migration fields and applies a fixed priority to supplied admission predicates. Both are implemented in `rtl/holotrade_admit.v`.

## 7.1 It is proved, not just written

**IN PLAIN ENGLISH: What formal verification is**

> Testing evaluates selected executions. Here, *formal equivalence* means a SAT solver found no input on which an encoded RTL circuit differs from its encoded reference. That is exhaustive for the stated finite input model; it is not proof that upstream predicates are truthful or that the reference captures all deployment requirements.

Two independently written implementations of the form — staged $\mathbb{F}_3$ operators and integer arithmetic with one final reduction — are equivalent over all $2^{16} = 65{,}536$ encoded address pairs. A second miter checks the full combinational admission and refusal-priority function over all $2^{25} = 33{,}554{,}432$ encoded inputs, including invalid-address handling and the seventh refusal code `BAD_ADDR`. PROVED

```
SAT proof finished - no model found: SUCCESS!
```

Equivalence is necessary and *not sufficient*: a reference can encode the wrong requirement. The software suite separately maps the 40 canonical points to the RTL representation and exhaustively checks all 1,600 ordered point pairs. Security predicate generation remains outside this proof boundary.

## 7.2 Current synthesis footprint

Synthesised to an iCE40 HX8K field-programmable gate array:

**49** × SB_LUT4 **6** × SB_CARRY MEASURED

This footprint covers address validation, form/adjacency, migration cost and the Boolean refusal-priority classifier. It excludes the cryptographic and stateful systems that produce `sig_ok`, `nonce_fresh` and `pins_ok`.

**What is deliberately not reported**

> This is Yosys iCE40 synthesis only. No place-and-route timing, clock frequency, power, latency or hardware measurement is reported; those are properties of a placed-and-routed design on a specific part.

### 7.3 The gate refuses rather than degrades

Seven refusal codes are included in the fixed priority function. The RTL classifies already-computed predicates; the full $2^{25}$ miter checks the code and priority for every encoded input.

| # | Refusal | Meaning |
|---|---|---|
| 1 | `BAD_SIG` | The upstream signature-verification predicate is false. |
| 2 | `REPLAY` | The upstream replay-store predicate reports a used nonce. |
| 3 | `WINDOW` | The supplied time-window predicate is false. |
| 4 | `PIN_DRIFT` | The supplied artifact-integrity predicate is false. |
| 5 | `NO_MAGIC` | Declared workload capability exceeds declared node capability. |
| 6 | `IN_SERVICE` | The node is unavailable for placement. |
| 7 | `BAD_ADDR` | One or both encoded addresses are not canonical valid points. |

There is no best-effort output in this classifier: it admits or emits one code. Deployment assurance still depends on authentic predicate sources, a trusted boot path and preventing bypass or FPGA reconfiguration.

# 8 What exists today

| Component | Status | Evidence |
|---|---|---|
| Cell geometry & adjacency | PROVED | exhaustive 40-point graph checks; SAT form equivalence |
| Six-term pricing engine | MEASURED | implemented decomposition and floor invariants |
| Two-sided balancer | MODELLED | paired 64-seed result reported with confidence intervals |
| Coherence & swap book | MODELLED | induced-edge objective; both modeled scores must rise |
| Recursive address metric | MODELLED | $40^n$ count; software bound $16n-14$; no physical product graph |
| Plan seal & admission policy | MEASURED | demo integrity seal; seven predicate-input refusal codes |
| Per-second metering | MEASURED | prototype prorates hourly-equivalent quotes by node-second |
| Contract templates | MODELLED | scenario pricing; no live market or legal contract |
| Trading terminal | MEASURED | browser prototype with seed data |
| RTL policy primitive | PROVED | form and full admission miters; synthesis 49 LUT4 + 6 carry |
| Physical photonic substrate | — | **does not exist**; not required by anything above |
| Real market data feeds | — | seed data; engines are feed-agnostic by construction |

## 8.1 The reproducibility contract

The repository checks the finite and implementation claims with:

```
npm test                                        finite checks and prototype contracts
npm run verify:rtl                              SAT-prove form and admission miters
node experiments/balancer_ab.js -summary        reproduce the paired 64-seed packet
```

The simulation uses a seeded pseudo-random generator. The frozen headline packet in `data/balancer_ab_64.json` uses 64 paired seeds, 220 nodes and 500 one-minute steps (8.333 simulated hours per arm); the same seed initializes each on/off pair.

**Two kinds of test, deliberately not mixed**

**Substrate facts** are exact finite mathematics with single correct answers. Any deviation is a bug, not a tuning question.

**Engine contracts** are implementation invariants: the floor is not breached, capability refusal has a reason, middle-entry deletion breaks the demo chain, and one quote interface accepts leaves and composites.

**Model experiments** depend on coefficients and are reported separately. A test that merely returns or recomputes a chosen constant establishes code consistency, not an external theorem or empirical result.

## 8.2 Two bugs the tests caught, and why that is the point

Both were found by the suite during development and both are the kind that survive casual review.

**A doubled topology score.** An earlier edge normalization used the wrong denominator. The corrected full-cell normalization is 100, while partial-basket outputs are now described accurately as induced-edge coherence rather than actual bisection.

**A non-monotone price curve.** The demand multiplier had a discontinuity at one of the seams between its discount, band and premium regions, creating a narrow range of utilisation where a machine got *cheaper by becoming busier*. That is directly arbitrageable: load the machine you are about to rent, then rent it at a discount. A monotonicity test caught it, and the fix is documented in the source beside the code it corrects.

# 9 Honest scope

This section exists because a document that only argues one direction is not worth the paper. Every claim here is sorted by what actually supports it.

**Exact and computed in this repository**

The $W(3,3)$ geometry — 40 points, 240 edges, 40 lines of 4 points each, diameter 2, bisection exactly 100 as the spectral bound met by the displayed explicit cut, and symmetry-group order 51,840. The $40^n$ recursive address count. The form miter over $2^{16}$ encoded pairs, the full admission miter over $2^{25}$ encoded inputs, and the 1,600-pair canonical-point cross-check. The two Hermitian ladder ceilings and their hole graphs; the 120-state coset graph and its SAT-equivalent coordinate/ROM circuits; the $15 \times 8$ fibre quotient with weighted matrix $4I + 2A(T(6))$; the square-zero rank-40 characteristic-two adjacency; the canonical $O_2(G)$ fibre partition; the 171-LUT affine relation compiler and its ROM miter; the 240-bit depth-20 reversible symplectic dilation; the 168-state Fano matched-action codec; the 20,160-state direct-product control chart; the 85-state polarity/sentinel control plane and its 216 five-circuits; and the stated full-normalizer stable-module identity and orientation/rollback no-gos.

These are statements about finite software objects and encodings. They do not establish physical topology, timing, security of upstream predicates or market performance.

**Modelled, and never presented as measured**

The fleet, wholesale prices, accumulated genomes, wear trajectories, order flow, partial-basket coherence, recursive distance, $9^t$ cost index, 58-reset Landauer calculation and contract valuations are models. This is a seeded simulation, not a production system with customers.

The modules expose feed interfaces. Real integration would still require units, timestamps, missing-data policy, authentication, settlement reconciliation, calibration and changes discovered during deployment.

**Asserted by the source programme, not by this product.** The $W(3,3)$–$E_8$ substrate programme makes claims about $W(3,3)$ being a candidate physical substrate for fundamental physics. **Holotrade does not depend on any of that being true.**

The cell graph and its adjacency rule are finite mathematics independent of the physics programme. Error-correction interpretation, thermodynamic accounting, recursive physical topology and quantum capability pricing add assumptions and are labeled as models here.

**The photonic hardware that would realise the quantum layer does not exist.** What runs here is ordinary classical software for the exchange and finite graph. The quantum field is a declared capability and illustrative price index, not a capability being offered.

## 9.1 The risks worth stating

- **Supply-side cold start.** A two-sided market needs machines and buyers. The prototype neither establishes safe lending nor makes physical joining free, and it does not create either side of the market.
- **The genome premium must be demonstrated, not argued.** The claim that a specialised core finishes in measurably fewer hours is plausible, grounded in existing practice, and *not yet measured on real workloads*. It is the single most important thing to validate, because it is where the durable margin lives. Until it is measured on production jobs, it is a hypothesis with a good mechanism.
- **Per-second settlement is an operational commitment.** Metering every running instance every second against a live grid price is real infrastructure with real failure modes, and the reconciliation problem when a meter disagrees with an invoice is not solved by having the right formula.
- **Regulatory surface.** Transferable forwards and options can create financial, commodities, securities, licensing, tax and consumer-protection issues. Product classification requires qualified counsel in every offered jurisdiction; this paper provides no legal conclusion.

# A Glossary

| Term | Meaning |
|---|---|
| Adjacency | Two graph vertices sharing an edge. Here it is determined by the symplectic predicate on canonical addresses. |
| Bisection | The minimum number of edges crossing any balanced partition of a graph. It is a topological quantity; physical bandwidth also needs link rates. |
| Clifford operations | Operations in stabilizer circuits that admit polynomial-time classical simulation under the Gottesman–Knill framework. Polynomial does not mean free. |
| Coherence | The prototype's normalized induced-edge basket score. It is not actual partial-graph bisection or measured bandwidth. |
| Content-addressed | Identified by a digest of content. Substitution is detectable only if a trusted verifier recomputes and authenticates the expected digest. |
| Correctable error | An error within the correction capability of the relevant ECC scheme. Counts are candidate reliability features, not deterministic failure predictions. |
| Derate | The gap between a machine's nameplate performance and what it actually delivers today, after wear. |
| Diameter | The worst-case number of hops between any two machines in a network. |
| Firecracker | Amazon's open-source microVM hypervisor, which runs their serverless products. |
| Gini coefficient | A number from 0 to 1 measuring how unevenly something is distributed. Used here for load across a fleet; lower is better. |
| Hash chain | A log whose entries commit to their contents and predecessor. Verification must recompute hashes and trust or externally anchor a chain head. |
| Hypervisor | Software that creates and isolates virtual machines on physical hardware. |
| Landauer limit | The thermodynamic minimum energy to erase one bit, about $kT \ln 2$. A physical law, not an engineering target. |
| MicroVM | A virtual machine with a reduced device model and separate guest kernel. Isolation still depends on implementation and deployment. |
| Mobility score | The modeled fraction of cell positions eligible under declared policy. It is not group orbit size or observed liquidity. |
| Node-second | One machine allocated for one second, charged at one 3,600th of the displayed hourly-equivalent rate plus itemised fees. |
| Non-Clifford / magic | Operations outside the Clifford stabilizer set. Their classical simulation cost depends on the circuit and algorithm; $9^t$ is this prototype's index. |
| Pin drift | An observed artifact no longer matching its expected digest. Production enforcement requires trusted resolution, hashing and attestation. |

| Term | Meaning |
|---|---|
| PUE | Power Usage Effectiveness: total facility power divided by computing power. A straight multiplier on the electricity bill. |
| Splice | Filling an unused modeled address slot with a local registry update. Physical integration is out of scope. |
| Symplectic form | The arithmetic expression on two canonical point representatives whose vanishing defines graph adjacency. It does not select a full route. |
| Thermal cycling | Repeated heating and cooling, one of several mechanisms that can contribute to hardware wear. |
| Weibull hazard | A parametric lifetime model. A fitted shape above 1 represents increasing hazard; the prototype coefficient is not field-calibrated. |

# B Selected constants and parameters

The graph constants below are exact for $W(3,3)$. The remaining rows are explicit inputs inherited from the source programme or chosen by the prototype; they are not forced engineering quantities.

| Value | Symbol | Status | Role |
|---|---|---|---|
| 3 | $q$ | exact | field order / address radix |
| 2 | $\lambda$ | exact | common neighbours of an adjacent pair |
| 4 | $\mu$ | exact | common neighbours of a non-adjacent pair |
| 12 | $k$ | exact | graph degree |
| 40 | $v$ | exact | vertices per complete cell |
| 240 | $\|E\|$ | exact | undirected edges per complete cell |
| 100 | — | exact | minimum balanced edge cut of the complete cell |
| 51,840 | $\|\mathrm{Aut}\|$ | exact | full graph-symmetry order used by the source model |
| 27/80 | — | source-model input | declared code-rate parameter; not a measured venue throughput cap |
| 1/10 | — | source-model input | contextual fraction; not production attestation evidence |
| 58 | — | conditional model | reset-record count used in the Landauer calculation |

# C References

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github.com/wilcompute/Holotrade | MIT | August 2026
Built on the $W(3,3)$–$E_8$ substrate programme.